Project Title

E-Commerce Sales Analysis and Performance Dashboard using Power BI

Introduction

The objective of this project is to analyse e-commerce sales data using Power BI and generate meaningful business insights through DAX calculations and interactive visualizations. The project focuses on evaluating sales performance, profitability, sales targets, and regional trends to support data-driven decision-making.

Dataset Description

The analysis was performed using three datasets:

DATASET	DESCRIPTION
List of Orders	Contains Order details, Customer Information, City, State, Sales amount and Profit.
Order Details	Contains category, sub-category, quantity, amount, and profit information.
Sales Target	Contains category-wise monthly sales targets.

DAX Calculations

The following calculated columns and measures were created:

Calculated Columns

Category Type

Revenue per Order

Sales Category

Measures

Order Count

Average Profit in Delhi

Year-to-Date (YTD) Sales

These calculations helped derive important business metrics for analysis and reporting.

Data Visualization and Analysis

The dashboard includes various visualizations to analyse sales performance and business trends effectively.

*Sales Target Achievement by Category: Used to compare actual sales against predefined sales targets and identify categories that met or missed expectations.

* Profit Margin by Sub-Category: Helps determine which sub-categories generate the highest profit margins and contribute most to business profitability.

* Monthly Sales Trend: Displays sales performance over different months, making it easier to identify growth patterns and seasonal variations.

* Profit vs Quantity Analysis: Examines the relationship between the quantity of products sold and the profit earned from those sales.

* Sales and Target KPI Cards: Provides a quick overview of key business metrics such as total sales and sales targets.

* Sales Performance Matrix: Enables comparison of sales performance across categories and different time periods.

* Geographical Sales Analysis: Visualizes sales distribution across cities and states to identify high-performing regions.

* Sales Distribution Tree map: Shows the contribution of each sub-category to overall sales, helping identify top-performing products.

* Order Count Analysis: Analysis the number of orders across different states to understand customer distribution and market reach.

Dashboard Features

Interactive and user-friendly dashboard

Clear visualization of sales performance

Business KPI monitoring through cards and measures

Regional sales analysis using maps

Trend analysis using line charts

Comparative analysis using matrix and scatter charts

Key Insights

Electronics category generated the highest revenue.

Certain sub-categories contributed significantly to overall profit.

Sales performance varied across different months and regions.

Comparison with sales targets helped identify performance gaps.

Geographic analysis revealed major sales-contributing cities and states.

Conclusion

This project successfully utilized Power BI, DAX calculations, and data visualization techniques to transform raw e-commerce data into actionable business insights. The dashboard enables effective monitoring of sales performance, profitability, and target achievement, helping organizations make informed business decisions and improve overall performance.